

Word Embeddings in Natural Language Processing

Word embeddings are numerical vector representations of words that capture semantic meaning. Unlike traditional representations such as Bag of Words or TF-IDF (which produce sparse vectors), word embeddings produce dense vectors where similar words have similar vector representations.

1. Why Word Embeddings?

Traditional text representations suffer from problems such as sparsity and lack of semantic meaning. Word embeddings solve these problems by representing each word as a dense vector in a continuous vector space. In this space, semantically similar words appear closer together.

2. Word2Vec

Word2Vec is one of the most popular methods for learning word embeddings. It was introduced by researchers at Google and trains a small neural network to learn vector representations of words. Word2Vec has two main architectures: Continuous Bag of Words (CBOW) and Skip-Gram.

3. Continuous Bag of Words (CBOW)

CBOW predicts a target word based on its surrounding context words. For example, given the sentence: "I love machine learning", if the target word is "love", the context words might be "I" and "machine". The model learns to predict the center word from these surrounding words. CBOW tends to be faster and works well for large datasets.

4. Skip-Gram Model

Skip-Gram does the opposite of CBOW. It predicts surrounding context words from a center word. For example, if the center word is "love", the model tries to predict words like "I" and "machine". Skip-Gram works particularly well for smaller datasets and is better at learning representations for rare words.

5. Negative Sampling

Training Word2Vec using a full Softmax layer can be computationally expensive because the model must compute probabilities across the entire vocabulary. Negative Sampling is an optimization technique that makes training much faster. Instead of updating weights for every word in the vocabulary, the model only updates weights for the correct context word and a few randomly selected incorrect words (called negative samples). This greatly reduces computation while still allowing the model to learn meaningful word embeddings.

6. Advantages of Word Embeddings

Advantage	Description
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Dense Representation	Embeddings are compact vectors rather than sparse matrices.
Semantic Similarity	Words with similar meanings have similar vectors.
Better NLP Performance	Improves many NLP tasks such as classification, search, and translation.

7. Applications

Word embeddings are widely used in modern NLP systems including chatbots, recommendation systems, machine translation, sentiment analysis, and search engines. They also form the foundation for modern transformer-based models such as BERT and GPT.